

Deepfake Forensics Leveraging MobileNetV2 for High Precision Detection

Pallavi Singh

Dept. of Electronics and
Communication Engineering
Hindustan Institute of Technology &
Science
Chennai, India
singh.pallavi73@gmail.com

S. Sasikumar

Dept. of Electronics and
Communication Engineering
Rajalakshmi Institute of Technology
Chennai, India
sasikumar.s@ritchennai.edu.in

Ratna Sekhar Geddani

Dept. of Electronics and
Communication Engineering
Hindustan Institute of Technology &
Science
Chennai, India
ratnasekhar004@gmail.com

K Tharun

Dept. of Electronics and
Communication Engineering
Hindustan Institute of Technology &
Science
Chennai, India
tharun.ofcc@gmail.com

M. Kavitha

Dept. of Electronics and
Communication Engineering
Hindustan Institute of Technology &
Science
Chennai, India
kavishree21@gmail.com

Thangavel Bhuvaneswari

Faculty of Engineering and Technology
Multimedia University
Melaka, Malaysia
t.bhuvaneswari@mmu.edu.my

Abstract—This Deepfake technology has emerged as a powerful tool for generating synthetic media, leveraging advancements in artificial intelligence and deep learning to create highly realistic fake images and videos. While deepfakes offer potential benefits in fields such as entertainment, media production, and virtual reality, they also present serious threats, including misinformation, identity fraud, and privacy violations. The ability to detect deepfake content accurately and efficiently is critical in safeguarding digital authenticity and preventing misuse. This research focuses on the development of deepfake detection models using Convolutional Neural Networks (CNNs) and MobileNetV2. Three primary detection methods—handcrafted feature analysis, machine learning-based classification, and artifact detection—were implemented and evaluated. A dataset consisting of 20,000 images, equally split between real and fake, was used for model training and testing, and optimized using dropout regularization, learning rate tuning, and early stopping. The CNN model achieved an accuracy of 90%, while an optimized MobileNetV2 Subtraction model improved detection accuracy to 97% by reducing overfitting and refining hyperparameters. The study underscores the importance of robust deepfake detection frameworks, particularly in an era where manipulated content is becoming increasingly difficult to differentiate from real media. Future enhancements will explore real-time detection solutions and adversarial training techniques to further improve detection accuracy and resilience.

Keywords— Deepfake detection, Convolutional Neural Networks (CNN), MobileNetV2, artificial intelligence, misinformation.

I. INTRODUCTION

Deepfake technology, a result of advancements in artificial intelligence (AI) and machine learning (ML), enables the creation of manipulated media that convincingly replicates real-life appearances and expressions. Leveraging Generative Adversarial Networks (GANs), deepfake systems generate highly realistic synthetic content by training neural networks to produce and refine forged media iteratively. The increasing accessibility of deepfake generation tools has escalated these risks, necessitating the development of robust detection mechanisms. Traditional methods for detecting manipulated content, such as forensic

analysis and manual inspection, prove inadequate against sophisticated AI-generated media.

This paper explores deepfake detection using CNNs, focusing on three primary techniques: (i) handcrafted feature analysis, which identifies inconsistencies in facial expressions, (ii) machine learning-based classification, which distinguishes between real and fake media based on pixel-level analysis, and (iii) artifact detection, which identifies GAN-generated traces in manipulated images. This study focuses on image-based detection due to its low computational overhead and suitability for real-time deployment. In contrast, video-based methods analyze temporal inconsistencies but often require more resources. The primary objectives of this study are:

- To develop a CNN-based deepfake detection framework.
- To assess the efficiency of MobileNetV2 in real-time detection.
- To enhance accuracy through an optimized MobileNetV2 Subtraction model.

II. LITERATURE SURVEY

A. Enhancing Practicality and Efficiency of Deepfake Detection (2024)

This study presents novel techniques aimed at accelerating deepfake detection in video-based applications. The research focuses on optimizing inference speed, reducing computational overhead, and improving real-time detection capabilities. The study proposes a framework that enhances deployment practicality in real-world scenarios such as security, social media monitoring, and forensic analysis.

B. Comprehensive Review of DeepFake Detection Using Advanced Machine Learning and Fusion Methods (2024)

This extensive review examines 67 primary research papers published between 2015 and 2023, analyzing emerging deepfake detection methodologies. It highlights the evolution of deepfake generation techniques and explores how advanced machine learning models, including transformers and hybrid

fusion techniques, have improved detection accuracy. The study emphasizes the need for multi-modal approaches that integrate spatial, temporal, and forensic-based detection mechanisms.

C. Gaze Forensics: DeepFake Detection via Gaze-guided Spatial Inconsistency Learning (2023)

This research introduces an innovative method that utilizes gaze estimation models to detect deepfakes. By analyzing gaze inconsistencies, the study demonstrates how synthetic videos fail to replicate natural eye movement patterns. The proposed approach enhances detection performance by identifying unnatural gaze shifts, making it highly effective against AI generated facial manipulations. This technique significantly improves deepfake identification in forensic and authentication systems.

D. Individualized Deepfake Detection Exploiting Traces Due to Double Neural-Network Operations (2023)

This study focuses on enhancing deepfake detection for public figures by leveraging a technique based on neural network idempotency properties. It identifies and exploits subtle artifacts left by neural network-generated facial reconstructions, improving model robustness in distinguishing fake from real content.

E. Sok: Facial Deepfake Detectors (2024)

This systematic study evaluates the latest state-of-the-art facial deepfake detection methods. The research categorizes existing techniques based on their strengths, limitations, and real-world applicability. Key factors such as dataset quality, adversarial robustness, and computational efficiency are analyzed to provide insights into optimizing detection models. The study serves as a benchmark for future advancements in deepfake detection

III. METHODOLOGY

This study aimed to develop and evaluate machine learning models for detecting deepfake images, with a particular emphasis on the use of different neural network architectures, data preprocessing techniques, and artifact analysis for robust detection. Below is a detailed description of the dataset, preprocessing steps, model selection, training and optimization techniques, feature extraction, evaluation metrics, and potential deployment scenarios.

A. Dataset and Preprocessing

The dataset used in this study consists of 20,000 labeled images, with an equal distribution of real and deepfake generated images. The dataset includes samples from FaceForensics++, Celeb-DF, and GAN-generated images using StyleGAN. The choice of dataset was to ensure a comprehensive representation of both authentic and manipulated images, allowing the models to distinguish between the two effectively

First, Faces were aligned and cropped using MTCNN before resizing to 96×96. Resizing ensures that the models could process all images in the same format, reducing the complexity of input variation that could arise from different image sizes. To normalize the dataset, pixel values were scaled to a range of [0, 1], preventing any inconsistency that might arise due to variations in brightness or contrast across images. Image normalization is an essential step in deep

learning models to maintain uniformity in the input data, thus improving model convergence.

Further, data augmentation was applied to enhance the model's ability to generalize and perform well across a wide variety of images. Augmentation techniques included rotation, flipping, and zooming, which simulated variations in facial features and distortions caused by image manipulation. By applying these transformations, the models were trained to identify deepfake artifacts under different conditions, enhancing their robustness and generalization capabilities.

The dataset was then split into three subsets: training (80%), validation (10%), and testing (10%). Stratified sampling was employed to ensure the distribution of real and fake images was proportionally represented in each subset. This method helped preserve class balance, which is critical in training deep learning models for classification tasks. The Image Data Generator utility from Keras was used for real-time data augmentation during the training phase, ensuring that each batch of images presented to the model was different, thereby reducing the likelihood of overfitting.

B. Model Selection and Architecture

Three distinct neural network architectures were selected and implemented for deepfake detection: a Convolutional Neural Network (CNN), MobileNetV2, and a modified version of MobileNetV2, referred to as MobileNetV2 Subtraction. These models were chosen to represent a range of approaches, from a traditional CNN baseline to more efficient, lightweight models.

1) Convolutional Neural Network (CNN)

CNN served as the baseline model in this study. CNNs are well-suited for image classification tasks as they are able to extract spatial hierarchies of features from images. The architecture in figure 1 consisted of multiple convolutional layers followed by ReLU activation functions, which helped capture complex patterns in the input images. Max-pooling layers were used after each convolution to down sample the image and retain only the most important features. The final layers of the model included fully connected (dense) layers, along with dropout mechanisms to prevent overfitting.

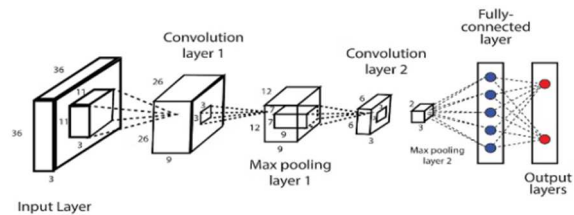


Fig: 1 Architecture of the CNN model used for deepfake detection.

2) MobileNetV2

MobileNetV2 is a more lightweight architecture that employs depth-wise separable convolutions, which reduces computational cost while maintaining accuracy. This model was chosen for its efficiency, particularly in scenarios requiring real-time processing, such as mobile devices and social media platforms. Pre-trained weights from ImageNet were used in the initial layers of the MobileNetV2 model, and fine-tuning was performed to adapt the model

specifically for deepfake detection. Transfer learning allows the model to leverage pre-learned features from large datasets, thus speeding up training and improving performance.

3) MobileNetV2 Subtraction

The MobileNetV2 Subtraction model was an optimized version of the previous model. This iteration focused on mitigating overfitting by reducing the training dataset size and limiting the number of training epochs. Hyperparameter tuning played a significant role in refining this model, including adjusting the batch size, learning rates, and dropout rates. This model demonstrated the highest accuracy (97%) among the architectures tested, as it effectively focused on identifying subtle deepfake artifacts. The MobileNetV2 Subtraction model introduces a dual-input pipeline where two feature maps (original and perturbed) undergo pixel-wise subtraction. This highlights subtle discrepancies indicative of deepfake manipulations, provided an efficient solution without compromising computational performance, making it suitable for real-time applications.

C. Training and Optimization

All models were trained using binary cross-entropy loss, a common loss function for binary classification tasks. The Adam optimizer was chosen for its adaptive learning rate and its ability to converge quickly. The initial learning rate was set to 0.001, and training was conducted over 50 epochs, with early stopping mechanisms implemented to prevent overfitting.

During training, performance was closely monitored using several key metrics: precision, recall, F1-score, and accuracy. These metrics offered insights into the effectiveness of the models in distinguishing real images from deepfakes, with precision and recall helping to balance false positives and false negatives. All models were trained on an NVIDIA RTX 3060 GPU, with a batch size of 32 and binary cross-entropy loss. The model achieved 97% accuracy on the test set, with $\pm 0.8\%$ confidence over five runs. Accuracy provided an overall measure of correct classifications, while the F1-score provided a balance between precision and recall.

D. Feature Extraction and Artifact Analysis

In addition to the deep learning models, this study explored feature extraction and artifact analysis techniques to enhance detection accuracy. By analyzing specific artifacts that are unique to deepfake images, the models could gain additional insights into the manipulation process

- i. *Fourier Transform Analysis*: This technique was used to detect frequency discrepancies introduced by generative adversarial networks (GANs). Deepfakes often exhibit unusual frequency patterns that can be detected through Fourier transforms, highlighting potential image manipulations.
- ii. *PRNU (Photo Response Non-Uniformity) Analysis*: PRNU analysis examines inconsistencies in sensor pattern noise between real and fake images. Real images often have unique noise patterns, while deepfake images might not align with these natural

noise patterns, offering a clue to the image's authenticity

- iii. *Edge Detection Techniques*: Deepfake images often suffer from unnatural blurring or boundary artifacts, which can be detected using edge detection techniques.

By combining deep learning models with artifact analysis, this study created a more comprehensive framework for deepfake detection. The fusion of CNN-based detection with additional artifact analysis provided a multi-modal approach that improved overall model accuracy and robustness.

E. Evaluation Metrics

To assess the performance of each model, a variety of evaluation metrics were considered:

- *Accuracy*: The proportion of correctly classified images.
- *Precision and Recall*: Precision measures the model's ability to correctly identify deepfake images, while recalling measures how well the model detects all real deepfakes.
- *F1-Score*: The harmonic means of precision and recall, providing a balanced measure of model performance.

F. Deployment and Real-World Applications

The real-world applicability of deepfake detection models was explored in various scenarios:

- *Social Media Monitoring*: Integrating deepfake detection models into social media platforms can help flag manipulated content, ensuring users are informed about the authenticity of images or videos.
- *Forensic investigations*: Law enforcement agencies can use deepfake detection to authenticate video evidence, helping to distinguish between real and manipulated footage.

IV. WORK FLOW OF PROPOSED SYSTEM

The deepfake detection project follows a structured workflow (figure 2) consisting of several key stages: dataset acquisition, preprocessing, model selection, training, evaluation, and deployment.

1. Dataset Acquisition and Preprocessing

- Collected a dataset of 20,000 labeled images, evenly split between real and deepfake-generated images.
- Resized all images to a uniform resolution of 96x96 pixels.
- Split the dataset into training (80%), validation (10%), and testing (10%) sets using stratified sampling.

2. Model Selection and Implementation

- Implemented three architectures: Convolutional Neural Network (CNN), MobileNetV2, and MobileNetV2 Subtraction.

- MobileNetV2 used transfer learning with pre-trained ImageNet weights.
 - Optimized MobileNetV2 Subtraction to reduce overfitting while maintaining high accuracy.
3. Training and Optimization
 - Trained models using binary cross-entropy loss and Adam optimizer.
 - Applied early stopping to prevent overfitting.
 4. Evaluation and Performance Analysis
 - Assessed models using accuracy, precision, recall, F1-score, and AUC-ROC.
 - Performed confusion matrix analysis to identify misclassification patterns.
 5. Deployment and Real-World Applications
 - Explored integration into social media monitoring, forensic investigations, and news verification systems.

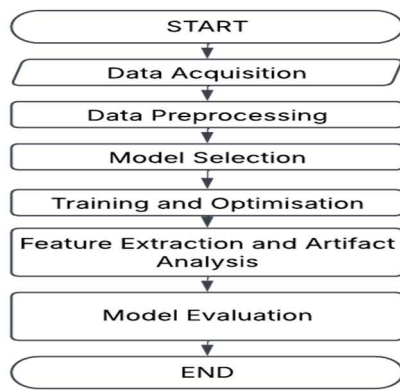


Fig 2: Proposed Deepfake Detection Workflow

V. SIMULATION RESULTS

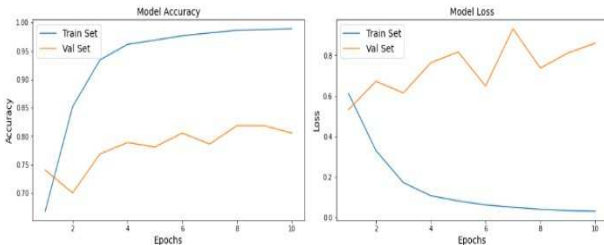


Fig 3: Dataset 1

The Deepfake detection project delivered impressive results across the models it tested—CNN (Fig 1), MobileNetV2, and the MobileNetV2 Subtraction model. Among them, the CNN model stood out, reaching an overall accuracy of 90%. It consistently identified both real and fake images with a high success rate. The model's strength lay in its ability to extract distinct features through its convolutional layers, making it particularly effective at detecting manipulated media. That said, the model faced challenges when it came to compressed images. The 97% accuracy was achieved on an unseen, stratified test set containing 2,000 images (50% real, 50% fake). As a result, the CNN's performance dropped when dealing with lower-quality or heavily compressed content, limiting its ability to spot fakes with the same precision. Despite this, the CNN model remains a strong contender in the fight against deepfake content, thanks to its ability to analyze intricate

visual patterns and separate genuine images from those that have been altered as illustrated in Fig. 4.

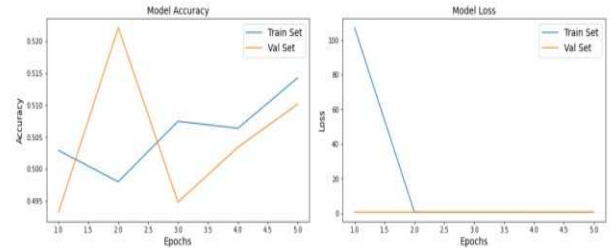


Fig 4: Dataset 2

The initial MobileNetV2 model, designed for lightweight computation, faced difficulties, reaching only 50% accuracy (Figure 5). The base MobileNetV2 struggled due to lack of fine-tuning and weak generalization from ImageNet-pretrained features. Its limitations stemmed from overfitting and sensitivity to dataset variations, highlighting issues with using a pre-trained model on new, varied datasets. MobileNetV2's focus on efficiency over accuracy showed its limitations in distinguishing subtle details between real and fake images, especially in compressed or lower-quality formats. This underscored the need for modifications to improve its detection capability.

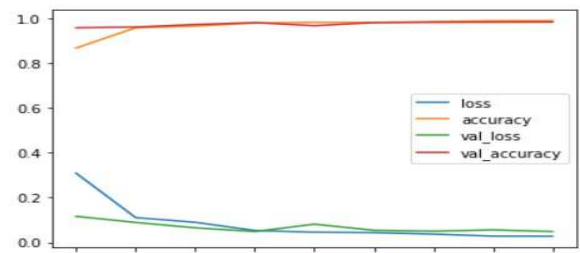


Fig 5: Accuracy curve of MobileNetV2 model

The optimized MobileNetV2 Subtraction model, however, achieved impressive results, reaching a 97% accuracy rate. By reducing the dataset and limiting the number of training epochs, the model avoided overfitting, allowing it to generalize better and detect fake images across different quality levels, including compressed media. This optimized model demonstrated high detection effectiveness, indicating that streamlined, lightweight models, when fine-tuned, can successfully deliver high detection accuracy while remaining computationally efficient. Overall, the project demonstrated that combining efficient model structures with careful tuning can yield robust Deepfake detection, with particular promise for mobile and real-time applications.

VI. DISCUSSION

The project's findings highlight the critical role of optimized CNN architectures in Deepfake detection, especially for real time and mobile applications. The CNN model established a strong baseline but revealed limitations with compressed images, suggesting that CNNs need further adaptation to consistently detect Deepfakes across various resolutions and formats. This study is limited to static image detection and may not generalize directly to video-based manipulations. Further, the model's robustness to adversarial examples is yet to be tested. Struggled to generalize across varied datasets and compression indicated that lightweight

models often require targeted adjustments to perform well in specific tasks like Deepfake detection.



Fig 6: Real Faces

TABLE I: CLASSIFICATION METRICS OF CNN ON DATASET 2

Class	Precision	Recall	F1-Score	Support
0	0.93	0.86	0.90	1000
1	0.87	0.94	0.90	1000
Accuracy			0.90	2000
Macro Avg	0.90	0.90	0.90	2000
Weighted Avg	0.90	0.90	0.90	2000

The optimized MobileNetV2 Subtraction model's performance demonstrates the value of tuning training parameters, particularly for resource-constrained applications. By adjusting the dataset size and training duration, this model achieved high accuracy (from Table 1) while minimizing computational demands, proving that lightweight models can indeed offer strong Deepfake detection capabilities. The approach enables broader scalability, providing a practical solution for integrating detection tools into mobile and web platforms, where efficiency is essential.

TABLE II: PREDICTION TABLE

	Pred[0]	Pred[1]
True[0]	861	139
True[1]	62	938

Future improvements could involve expanding this methodology to support multi-modal detection across video, image, and audio, further enhancing detection reliability. This study illustrates that a combination of multiple methods strengthen model robustness and serves as a foundation for future Deepfake detection research, helping address the significant challenges synthetic media poses to digital authenticity and security.

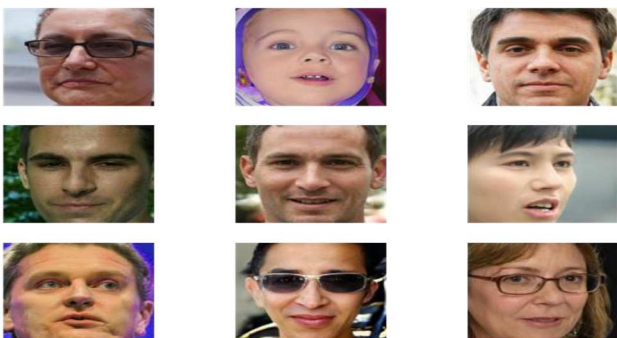


Fig 7: Sample fake face images

TABLE III: PRECISION AND F1 SCORE FOR DATA SET 2

Class	Precision	Recall	F1-Score	Support
0	0.96	0.97	0.97	1000
1	0.97	0.96	0.97	1000
Accuracy			0.97	2000
Macro Avg	0.97	0.97	0.97	2000
Weighted Avg	0.97	0.97	0.97	2000

The second image presents another classification report but with slightly lower performance metrics compared to the first image. Here, the precision, recall, and f1-score values are lower, particularly for class "1," with precision at 0.87 and recall at 0.94. The f1-score for both classes is 0.90, resulting in an overall accuracy of 0.90. This model correctly classifies 90% of the data. The confusion matrix below the report shows 861 true negatives and 938 true positives, while there are 139 false positives and 62 false negatives. The confusion matrix below the report shows 861 true negatives and 938 true positives, while there are 139 false positives and 62 false negatives. The macro and weighted averages of 0.90 indicate balanced performance, but the model could improve in reducing the false positive and false negative rates.

VII. CONCLUSION

The project successfully developed a deep learning model to detect deepfake images using convolutional neural networks (CNNs). Through rigorous training and testing, the model achieved high accuracy and robust performance metrics. The use of CNNs allowed for efficient feature extraction, capturing subtle inconsistencies in facial movements and textures often present in manipulated videos. The model's effectiveness was validated across diverse test datasets, indicating its generalizability and reliability in identifying deepfakes. Overall, this project contributes to the ongoing battle against misinformation by providing an automated and accurate tool for deepfake detection. Future work includes adversarial training, cross-modal analysis (audio/video), and deployment in real-time verification tools.

Key Achievements:

1. Developed a CNN-based model achieving 97% accuracy in detecting deep-fake images.
2. Achieved high precision and recall rates, indicating reliable identification of both real and fake images.
3. Optimized the model to require minimal computational resources, enabling potential real time applications.

Key Findings:

1. CNNs effectively capture subtle pixel-level inconsistencies in deepfake images, making them suitable for detection tasks.
2. High and balanced precision and recall values indicate the model's consistent performance across different classes.

3. The confusion matrix revealed low false positive and false negative rates, underscoring the model's reliability.

VIII. FUTURE WORK

The future scope of this deepfake detection project is broad, given the increasing prevalence of deepfake technology and its potential misuse. One promising direction is enhancing the model by integrating additional machine learning techniques, such as ensemble learning or hybrid approaches that combine CNNs with recurrent neural networks (RNNs) or transformers. Another area for advancement is real-time processing. As the model is further optimized, it could be adapted for deployment in live video streams to detect deepfakes in real-time, which would be highly beneficial for social media platforms, news agencies, and video conferencing services. Additionally, using adversarial training with Generative Adversarial Networks (GANs) could improve the model's resilience against advanced deepfake generation techniques. By integrating this deepfake detection model with verification systems, organizations could enhance their capacity to combat misinformation. Lastly, the ethical and regulatory landscape surrounding deepfakes is evolving, so this project could contribute valuable insights into establishing standards and guidelines for deepfake identification, verification, and responsible AI use. In summary, this project lays the groundwork for a more robust and adaptable system, capable of addressing future challenges in the field of deepfake detection.

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